# Part –III

# PRINCIPLES OF ELECTRONIC COMMUNICATION

## Chapter -7: Fundamentals of Analog Communication

Communication is the process of transferring information from one point to another. Based on the type of the signal transmitted, communication can be analog or digital. The analog communication involves transmitting of an analog signal such as music, video etc. from source to destination. In electronic communication, some kind of modulation of a known signal called carrier with the information or message signal before transmission is performed. In modulation one of the parameters of carrier such as amplitude, frequency or phase is varied in accordance with the instantaneous amplitude of the message signal. Depending on the parameter that is varied, there are three basic types of modulation called amplitude, frequency and phase modulation.

**Module – 1: Amplitude Modulation**

**Learning Outcomes:**

At the end of this module, students will be able to:

1. Explain the principle of electronic communication using a block diagram.
2. Define modulation and discuss the need for modulation.
3. Explain amplitude modulation using suitable waveforms and define modulation index.
4. Draw the spectrum of AM DSB signal, identify sidebands and estimate bandwidth.
5. Derive expression for power content of AM signal.
6. List and describe different types of AM signal and compare them.
7. Explain the principle of AM demodulation process.
8. Explain with suitable block diagram the principle of Super- heterodyne receiver.

### 7.1.1 Principles of Electronic Communication

In a broad sense, the term electronic communication refers to the sending, receiving and processing of information by electronic means. It can also be defined as the process of transmitting the information or signal from one point known as the source to another point known as the destination. Information can be continuous such as speech, music, image, picture etc. or discrete signals like data from computer etc.

The block diagram of basic communication system is shown in Figure 7.1.1.

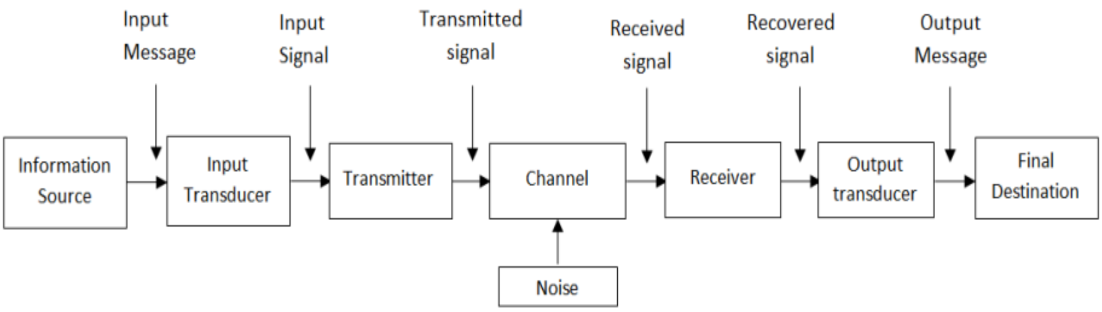



Figure 7.1.1: Block Diagram of Electronic Communication System

**a. Information Source:** The first stage of communication system is the information source because a communication system transmits information from an information source to the destination. The physical form of information is represented by a message that is originated by an information source. The examples of message are voice, live scenes, music, image, written text and e-mail etc.

**b. Input/output transducer:** The input transducer converts physical quantity (non-electrical) to an electrical signal. This electrical signal is called as baseband signal/ message signal. For example, voice is converted to electrical signal using microphone. Similarly at the destination, output transducer is used to convert electrical signal back to physical quantity. For example a loudspeaker is used to convert electrical signal back to voice. Likewise for video, the input transducer can be camera and output transducer can be any picture display unit such as Cathode Ray Tube (CRT) or Liquid Crystal Display (LCD).

**c. Transmitter:** The baseband signal generated by the transducer may not be in the form suitable for the transmission. Hence some kind of processing and signal conditioning is required to make it suitable for transmission. The transmitter section processes the signal before transmission, which mainly consists of filters, amplifiers, modulator and transmitting antenna (for wireless transmission). The filter is used to eliminate the unwanted component of signal generated by the transducer. The desired signal is further amplified to the required level using amplifier. The baseband signal is applied to the modulator, which translates the baseband signal from its low frequency to high frequency, makes it suitable for the transmission in the chosen environment. The modulated signal is then transmitted through a transmission channel.

**d. Transmission channel**: It is a medium over which the electronic signal is transmitted from one point to another. The type and characteristics of the channel along with the noise power decides the transmitter and receiver selection and design and hence the cost of the communication system. The communication medium can be either wired or wireless. An example for wired communication is telephony, where a pair of physical wires is running parallel between transmitter and receiver. Now-a-days optical fibers are used in between transmitter and receiver in which light carries the information. Similarly an example for wireless communication is radio communication, where free space is used as a transmission channel. Antennas are used to couple the signal to and from the channel to the communication system.

**e. Noise:** It is a random, undesirable electrical energy that interferes with the transmitted signal. Noise is a highly undesirable part of the communication system which must be minimized. The noise introduced in the transmission medium is called external noise and the noise introduced in the transmission and reception equipment is called as internal noise. The main cause of internal noise is the thermal agitation of atoms and electrons of the electronic components used in the equipment.

**f. Receiver:** The receiver block mainly consists of receiving antenna, filter, demodulator and amplifier. The signal received from receiving antenna is filtered and desired signal is amplified. It is further demodulated to get back the information signal. Finally an output transducer is employed to convert back the information in electrical form to physical form.

### 7.1.2 Need for Modulation

In wireless communication, the free space is the transmission medium through which electromagnetic waves carrying information propagates. A transmitting antenna at the transmitter radiates energy into the free space and it is received at the receiver using receiving antenna. The original signal which is called base band signal is not suitable for direct transmission/radiation over a long distance and hence modulation is usually performed.

Modulation is a process of varying some of the characteristics of high frequency carrier wave in accordance with the instantaneous amplitude of base band signal. After modulation the baseband signal of low frequency is transferred to the high frequency carrier, which carries information in the form of some variations. The three parameters of a sinusoidal carrier that can be varied are: amplitude, phase and frequency. A given modulation scheme can result in the variation of one or more of these parameters.

1. **Modulation for long distance communication:** According to electromagnetic theory, the amount of radiation emitted from an antenna depends on the frequency of the signal current supplied to it. For message signals of low frequency, radiation is poor and they get highly attenuated when transmitted over a longer distance. Therefore modulation is necessary to effectively increase the frequency of the signal to be radiated and thus increases the distance over which signal can be transmitted faithfully.
2. **Modulation to reduce the height of the antenna:** The height of the antenna required to transmit and receive radio waves is a function of wavelength of the signal. Since wavelength is inversely proportional to frequency for message signal having low frequency, λ is high and hence the height of the antenna required for transmission is very large and impractical. Therefore high frequency carrier is used to transmit the information which requires antenna of lesser height.
3. **Modulation for multiplexing:** Multiplexing is a method of transmitting more than one information signal simultaneously over a single channel. This allows several users to use the same channel if they are assigned with different carrier frequencies. Therefore it allows the maximum possible utilization of the available bandwidth of the system.

*Self- test:*

1. List the key components required for electronic communication.
2. List the basic functions of radio transmitter and the corresponding functions of the receiver.
3. Signals travel in air as --------------
4. ---------- is another name for simplex communication and ----- is another name for duplex communication.
5. Signals travel over copper cable as-----
6. Signals travel over fiber optic cable as-----

### 7.1.3 Amplitude Modulation (AM)

AM is defined as the process of varying the amplitude of the carrier wave proportional to the instantaneous amplitude of modulating signal. In practice, the carrier may be higher frequency while modulating signal may be lower frequency. Figure 7.1.2 shows the modulating, carrier and amplitude modulated wave. Here the modulating signal considered is multi-tone in nature.

**Note:**

- A multi-tone signal is the superposition of several sine waves or tones or frequency components, whereas single-tone signal has only one frequency component in it.

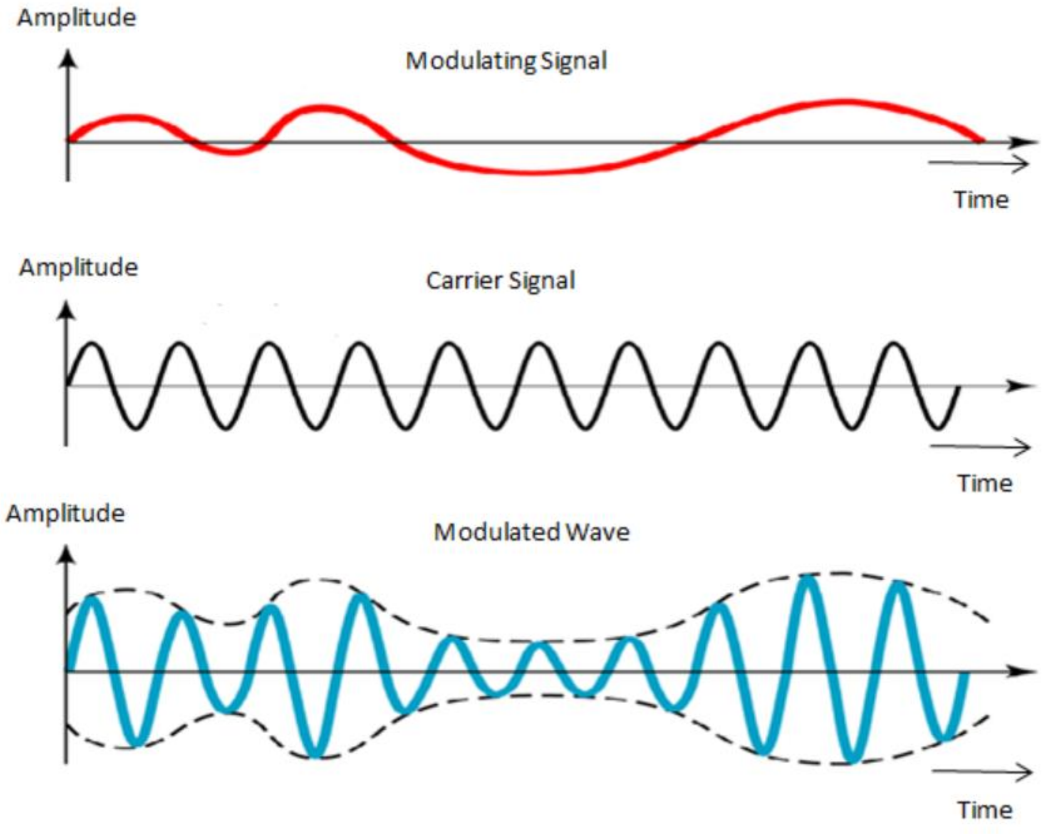



Figure 7.1.2: Waveforms of amplitude modulation

### *7.1.4 Time domain analysis*

The Figure 7.1.3 shows the time domain representation of the AM wave.

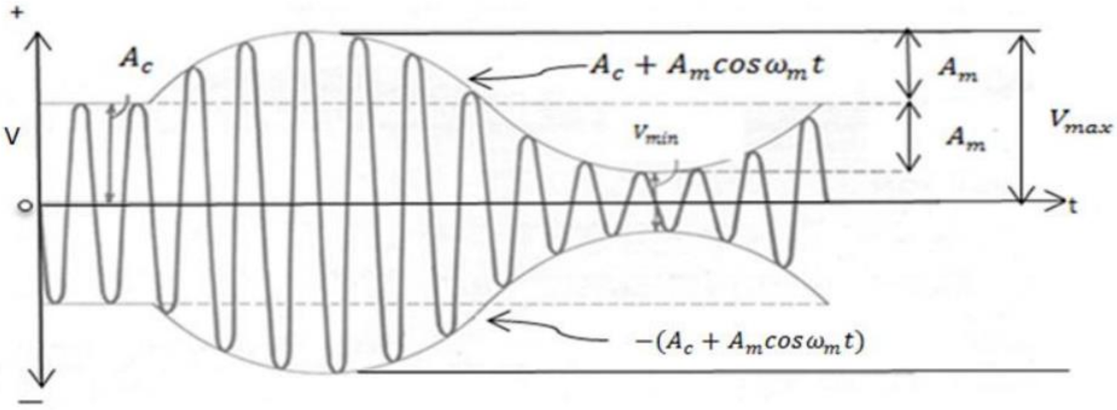



Figure 7.1.3: Time domain representation of AM wave

Let the equation of carrier signal be $c(t) = A_c\cos(2\pi f_c t)$, where $A_c$ is the peak amplitude of carrier signal is and $f_c$ is the frequency of the carrier signal.

Let the equation of modulating signal be $m(t) = A_m \cos(2\pi f_m t)$, where $A_m$ is the peak amplitude of modulating signal is and $f_m$ is the frequency of the modulating signal.

Then by the definition of AM:

$$V_{AM}(t) = [A_c + A_m \cos(2\pi f_m t)]\cos(2\pi f_c t) \qquad (7.1.1)$$

$$= A_c \cos(2\pi f_c t) + A_m \cos(2\pi f_m t)\ \cos(2\pi f_c t)$$

$$= A_c \cos(2\pi f_c t) + \frac{A_m}{2}\{\cos[2\pi(f_c + f_m)t] + \cos[2\pi(f_c - f_m)t]\}$$

$$V_{AM}(t) =$$

$$A_c \cos(2\pi f_c t) + \frac{mA_c}{2}\{\cos[2\pi(f_c + f_m)t] + \cos[2\pi(f_c - f_m)t]\} \qquad (7.1.2)$$

Where, ‘m’ is the modulation index of AM signal which is defined as ratio of the amplitude of modulating signal to that of carrier signal i.e.$A_m/A_c$ . The significance of modulation index is, it decides the depth of modulation. If it is less than one, then AM signal is known as under modulated signal. If it is more than one, then AM signal is known as over modulated signal. If it is equal to one, then AM signal is known as perfect modulated signal. To obtain the original information, modulation index should always be less than or equal to one. The effect of modulation index on AM wave is illustrated in Figure 7.1.4

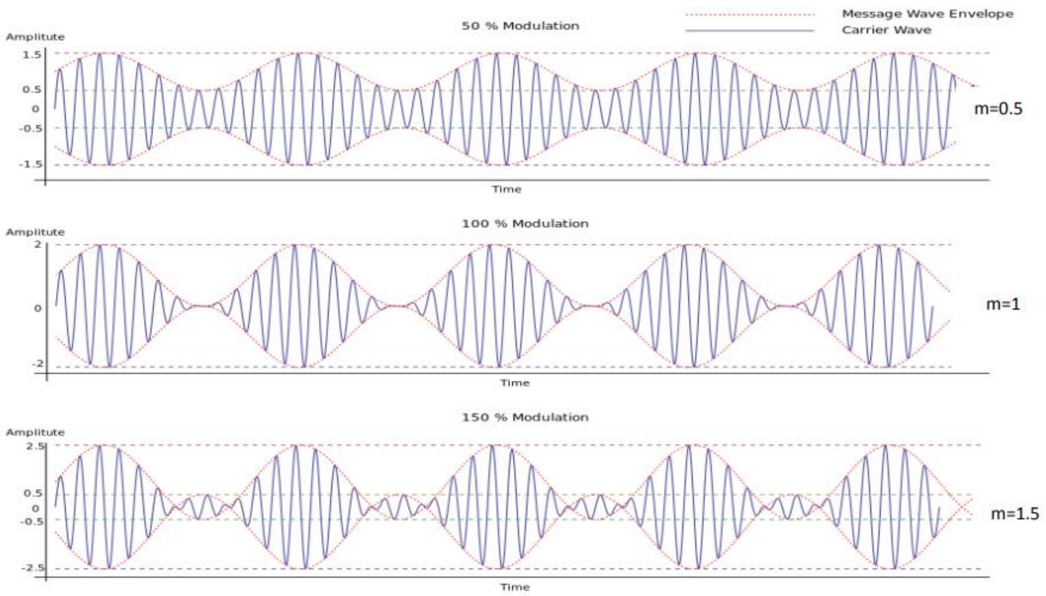



Figure 7.1.4: Effect of modulation index on AM waveform

### 7.1.5 Spectrum of AM signal

Another way of representing the AM signal is in the frequency domain known as frequency spectrum, which shows all the frequency components present in a signal. It is the plot of frequency versus their respective amplitudes. Spectrum of AM signal is shown in Figure 7.1.5.

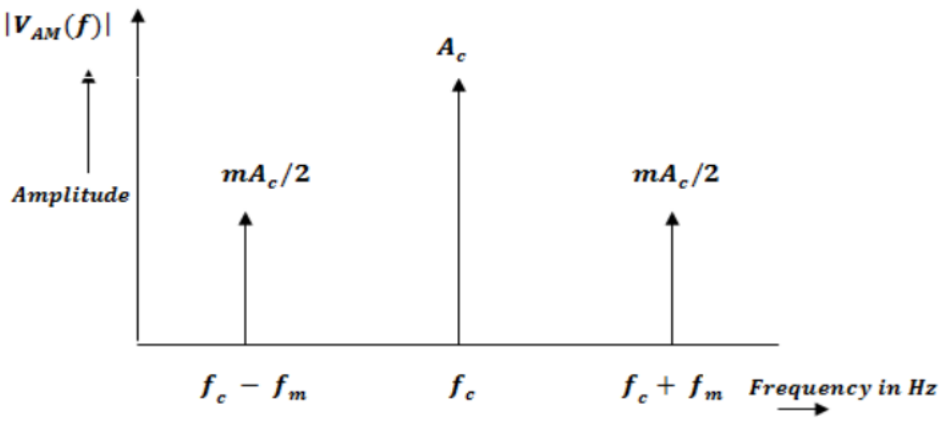



Figure 7.1.5: Frequency Spectrum of AM signal

As shown in Figure 7.1.5, the spectrum of AM consists of three frequency components, one at $f_c$ and other two at $f_c + f_m$ and $f_c - f_m$ respectively. The frequencies $f_c + f_m$ and $f_c - f_m$ are known as sideband frequencies i.e. $f_c + f_m$ is Upper Side Band (USB) and $f_c - f_m$ is the Lower Side Band (LSB). For this reason this is called AM DSB (Amplitude Modulation with Double Side Band) system. The difference between the two side band frequencies is defined as bandwidth of AM signal. Therefore the bandwidth of AM signal is $2f_m$.

*Self -test:*

1. What is a spectrum? What are all the information obtained from the spectrum of AM signal?
2. Imagine that there is a signal with two frequency components, $f_1$ and $f_2$ with $f_2 > f_1$. If they modulate a carrier with frequency $f_c$,

- Plot the frequency spectrum of the modulated signal.
- What is the bandwidth required for such a signal?

### 7.1.6 Power Content of AM

The total power (PT) of AM signal is given by

$$P_T = P_C + P_{USB} + P_{LSB} \tag{7.1.3}$$

Where, $P_C$ is the carrier power, $P_{LSB}$ and $P_{USB}$ are the side band signal powers.

$$P_T = \frac{A_c^2}{2R} + \frac{m^2 A_c^2}{8R} + \frac{m^2 A_c^2}{8R} \tag{7.1.4}$$

$$= \frac{A_c^2}{2R}\left\{1 + \frac{m^2}{2}\right\} \tag{7.1.5}$$

Where R is the load resistance.
Thus,

$$P_T = P_C\left\{1 + \frac{m^2}{2}\right\} \tag{7.1.6}$$

For 100% modulation, $m = 1$,
Therefore, $P_T = P_C\left\{1 + \frac{1}{2}\right\} = \frac{3}{2}P_C$

$P_C = \frac{2}{3}P_T$ or

$$P_C = 66.67\% P_T \tag{7.1.7}$$

i.e. 66.67% of total power is carried by the carrier and only 33.33% of total power is available in the sidebands. As the information is available only in the sidebands, and carrier does not in any way contribute to the information, 66.67 % of power is wasted if AM DSB with full carrier is used.

• **Modulation by several sine waves**:

When the modulating signal consists of several sine waves,

$$m(t) = A_{m1}\cos(2\pi f_{m1}t) + A_{m2}\cos(2\pi f_{m2}t) + \cdots$$

The overall modulation index will be,

$$m_t = \sqrt{\frac{{A_{m1}}^2 + {A_{m2}}^2 + \cdots}{{A_c}^2}} \qquad (7.1.8)$$

Or

$$m_t = \sqrt{{m_1}^2 + {m_2}^2 + \cdots} \qquad (7.1.9)$$

Therefore, the total power is,

$$P_T = P_C\left\{1 + \frac{{m_t}^2}{2}\right\} \qquad (7.1.10)$$

**Example Problem 1:**

1. An audio signal of $10sin\,(2\pi 1000t)$ volts amplitude modulates a carrier of $40sin\,(2\pi 2000t)$ volts. Find
   i. Modulation index
   ii. Sideband frequencies
   iii. Bandwidth
   iv. Total power delivered if $R_L = 1k\Omega$
   v. Amplitude of each side band components

**Solution:**

i. Modulation index: $m = \frac{A_m}{A_C} = \frac{10}{40} = 0.25$

ii. Sideband frequencies :
Upper side band = $f_C + f_m = 3000Hz$
Lower side band = $f_C - f_m = 1000Hz$

iii. Bandwidth = $2f_m = 2kHz$

iv. Total power delivered:
$P_T = \frac{A_C^2}{2R}\left(1 + \frac{m^2}{2}\right) = \frac{1600}{2000}\left(1 + \frac{(0.25)2}{2}\right) = 0.825$ Watts

v. Amplitude of each sideband = $m\frac{A_C}{2} = 0.25 * \frac{40}{2} = 5V$.

**Example Problem 2:**

Certain AM transmitter radiates 9 kW of power with carrier unmodulated and 10.125kW of power when carrier is sinusoidally modulated. Calculate the modulation index. If another sine wave corresponding to 40% modulation is transmitted simultaneously, determine the total power radiated.

**Solution:**

i. Given: $P_C = 9\text{kW}$

$P_T = 10.125\text{kW}$

$$P_T = P_C \left\{ 1 + \frac{m^2}{2} \right\}$$

$$m = \sqrt{2\left(\frac{P_T}{P_C} - 1\right)} = 0.5.$$

ii. $m_1 = 0.5$, $m_2 = 0.4$, $P_C = 9\text{kW}$.

$$m_t = \sqrt{m_1^2 + m_2^2} = 0.64.$$

$$P_T = P_C \left\{ 1 + \frac{m_t^2}{2} \right\} = 10.84\text{kW}$$

***Exercises:***

1. Show that modulation index $= \dfrac{V_{MAX} - V_{MIN}}{V_{MAX} + V_{MIN}}$ , where $V_{MAX}$ and $V_{MIN}$ are maximum and minimum voltage values of the envelope of AM signal.
2. A 360W carrier is simultaneously modulated by two audio waves with percentage modulation of 55 and 65 respectively. Find the modulation index, total power radiated and power in each sideband. Assume $R_L = 1\Omega$. [Ans: $m_t = 0.85$, $P_T = 490\text{W}$, $P_{USB} = P_{LSB} = 65\text{W}$].
3. A broadcast AM transmitter radiates 10kW when the modulation percentage is 60. How much of this is the carrier power? [Ans: $P_c = 8.47$ kW]

### 7.1.7 Different Types of AM Signals

The basic form of AM signal is called Double Side Band with Carrier (AM-DSB), where it contains two side bands namely, LSB, USB and the carrier signal in its unmodulated form. The transmission efficiency of the AM-DSB transmitter is very poor because the maximum efficiency that can be achieved is only 33.33%. The remaining 66.67% power is lost in the unmodulated carrier signal, which does not carry any useful information. The two side bands carry similar information and together they carry 33.33% of the transmitted power. Significant amount of transmitting power can be saved if the AM signal is modified before its transmission, accordingly several forms of AM signals have been developed which are

explained below.

1. **Double Side Band- Suppressed Carrier (DSB-SC) Signal:** It is the first modified version of AM-DSB signal. In a AM-DSB signal carrier does not contain any useful information. Therefore, the two side bands are transmitted by suppressing the carrier signal leading to a savings of 66.67% of the total transmitting power. This modified AM signal is called as Double Side Band Suppressed Carrier (DSB-SC) signal.
2. **Single Side Band (SSB) Signal:** The two side bands transmitted in DSB-SC signal are identical and carry similar information. If only one side band is transmitted, even then the information will be satisfactorily communicated. This gives rise to another form of AM known as Single Side Band (SSB) modulation. Thus in SSB, only one side band is transmitted by suppressing the other side band and the carrier signal. This also results in reduced transmitter power compared to DSB SC. The main disadvantage of AM-SSB is, it requires complex receiver.
3. **Vestigial Sideband (VSB) Signal:** It is another form of AM modulation, where one sideband is completely present, and a part (vestige) of other sideband is retained. If the carrier signal is transmitted along with the side bands, then the recovery of the baseband signal becomes easier. This also reduces the complexity of the receiver circuit and which in turn reduces its cost.

**Note:**

- AM-DSB signal is also referred as DSB with full carrier (DSB-FC) and the AM without carrier is called DSB with suppressed carrier (DSB-SC) signal.
- Watch this Video for animation of amplitude modulated output with different modulation indices (DSB): http://www.youtube.com/watch?v=1wUjLWNgqMs

Table 7.1.1 Comparision of different types of AM for m=1

| AM Scheme | Bandwidth | Carrier power | Side band power | % of power saving as compared to AM-DSB | Typical Applications |
|---|---|---|---|---|---|
| AM-DSB | $2f_m$ | 66.67% | 33.33% | NIL | AM radio broadcast |
| DSB-SC | $2f_m$ | NIL | 33.33% | 66.67% | Non-commercial systems |
| SSB | $f_m$ | NIL | 16.67% | 83.33% | Carrier telephony systems, military applications |

*Self-test:*

1. Compare different types of Modulation techniques based on a) bandwidth required and b) power content of the signal?
2. Calculate the percentage of power saving when the carrier and one of the side bands are suppressed in an AM wave modulated to a depth of (a) 100% (b) 50%
3. Explain why AM-DSB is preferred for commercial radio broadcasting?

### 7.1.8 Detection of AM Signal

Amplitude modulation or AM is one of the most straight forward ways of modulating a radio signal or carrier. In the process of demodulation (detection), the audio signal is removed from the radio carrier in the receiver. Demodulation is a process of recovering the original base band signal (information) from the modulated signal. The simple and highly effective method for demodulation is by using envelope detector.

Envelope detector produces an output signal that follows the input signal waveform exactly. Figure 7.1.6 shows the circuit diagram of an envelope detector that consists of a diode and a resistor-capacitor filter.

This is essentially a half wave rectifier with filter circuit, which allows only half of the alternating waveform through. The capacitor bypasses the high frequency carrier component and allows low frequency message signal to go to the output. This demodulator is applicable only for AM-DSB and the main advantage of this form of AM demodulator is that it is very simple and cost effective.

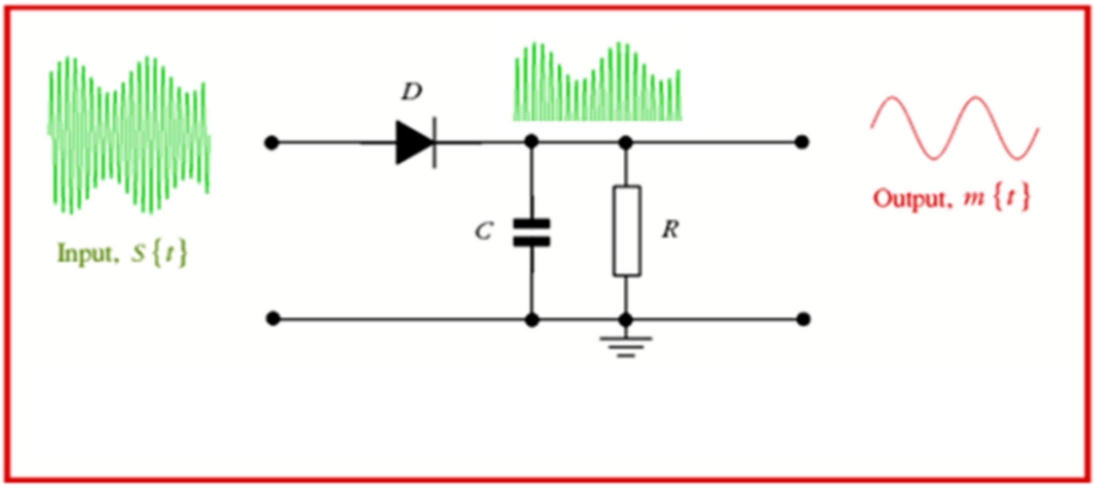



Figure 7.1.6: Envelope Detector

### 7.1.9 Super-heterodyne principle of AM Detection

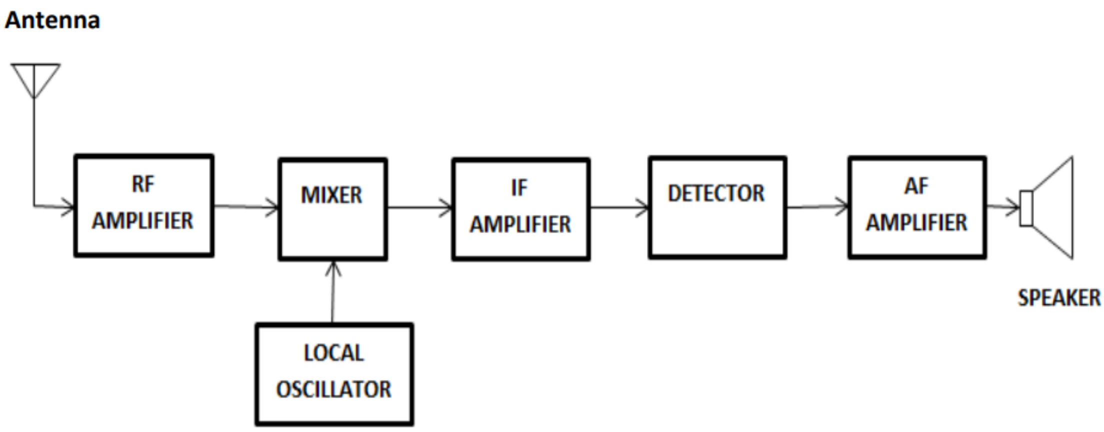



Figure 7.1.7: Block diagram of AM Super-heterodyne Receiver

There are a great variety of receivers in communication systems based on the requirements such as the modulation scheme, the operating frequency and its range. One of them is super heterodyne type, which uses frequency mixing or heterodyning to convert a received signal to get a fixed intermediate frequency

(IF). This allows the processing of signal easier as the circuits after IF needs to be designed for narrow band of frequency. The functional block diagram is shown in Figure 7.1.7.

The received signal from the antenna is amplified by the RF amplifier and is fed to the mixer stage, which performs the heterodyning of the incoming signal with the local oscillator signal to produce the sum and the difference of those two frequencies. The IF amplifier will be tuned to the difference frequency as it is smaller among the two, and is known as the intermediate frequency (IF). A typical value of IF for an AM communication receivers is 455 KHz. The difference frequency is at a lower frequency than either the RF input or oscillator frequencies.

Once the IF stage/stages have amplified the intermediate frequency to a sufficient level, it is fed to the detector. The detector is used to demodulate the signal and to get back the message. The detector stage consists of a rectifying device and filter, which respond only to the amplitude variations of the IF signal. This develops an output voltage varying at an audio frequency rate. The output from the detector is further amplified in the audio amplifier and is used to drive a speaker or earphones.

**Summary**

1. Basic principle of electronic communication, which has essentially three components:
   a) Transmitter, b) Receiver and c) Channel

2. Definition of modulation, which is nothing but varying some parameter of a known signal called carrier in accordance with the amplitude of the message signal called modulating signal.
3. Modulation is necessary for the following reasons:
   a) Ease of radiation b) efficient transmission and c) supporting multiplexing

4. To draw the waveforms for amplitude modulated signal with respect to the chosen modulating and carrier signals.
5. Modulation index gives the depth of modulation or the extent to which the carrier is modulated by the signal and is given by $\frac{A_m}{A_c}$
6. Draw the spectrum of AM-DSB signal for a single tone modulation and identify two sidebands and the carrier. The bandwidth of AM DSB is given by $2f_m$, where $f_m$ is the maximum frequency component of the modulating signal.
7. The power content of AM DSB is given by
   $$P_T = P_C\left\{1 + \frac{m^2}{2}\right\}$$
8. The different types of AM signal are AM DSB with carrier, DSB SC, SSB SC, SSB with carrier and VSB.
9. AM DSB can be demodulated by relatively simple process of envelope detection.
10. One of the popular AM reception method is called super heterodyne principle, where in the input RF signal is translated to an IF signal by mixing or beating it with the output of local oscillator. Since the local oscillator frequency is maintained above the incoming signal frequency it is called Super- heterodyning.

**Module – 2: Frequency Modulation**

In the previous module, we have learnt amplitude modulation for transmission of analog signals. Here, we will discuss about another type of analog modulation called Frequency Modulation (FM) where the frequency of the carrier is varied in accordance with the instantaneous amplitude of the modulating signal.

**Learning Outcomes:**

At the end of this module, students will be able to:

1. Define Frequency Modulation(FM)
2. Explain frequency modulation using suitable waveforms and define modulation index.
3. Illustrate the concept of FM graphically
4. Write the time domain equation for FM and determine the bandwidth of FM using Carson rule
5. Distinguish between AM &FM.

### 7.2.1 Frequency Modulation

Frequency Modulation is defined as a process of altering the frequency of the carrier signal with respect to the instantaneous amplitude of the modulating signal. It is illustrated in Figure 7.2.1.

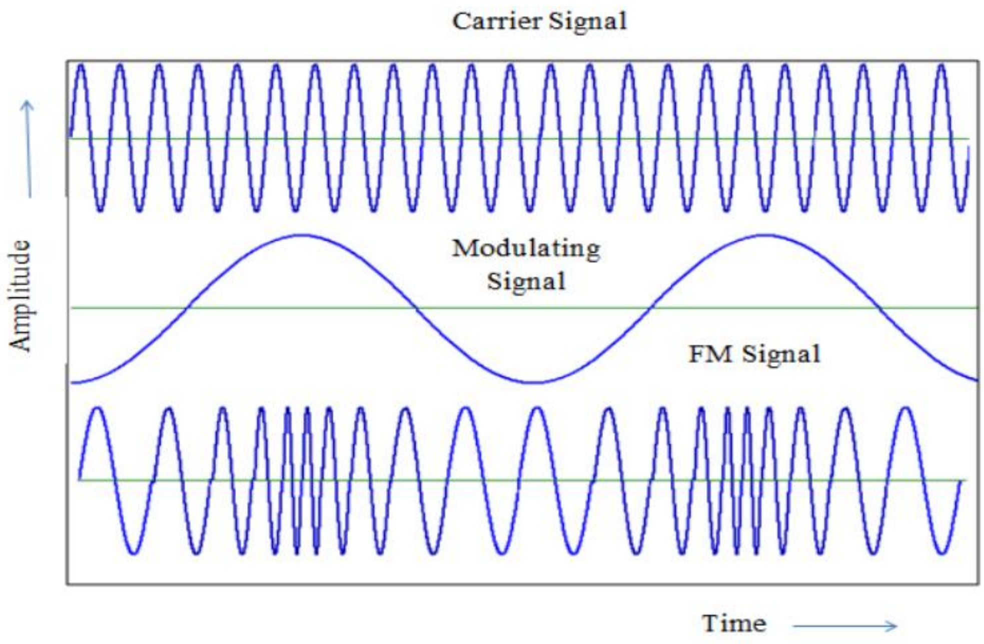



Fig.7.2.1: Frequency Modulation waveforms

### 7.2.2 Time domain analysis of FM signal

- Time domain analysis of FM signal

From definition, the frequency of the FM modulated signal is,

$$f_{FM} = f_c + K_f m(t) \tag{7.2.1}$$

Where $K_f$ is the frequency sensitivity, $f_c$ is the carrier frequency and $m(t)$ is the message signal or modulating signal.

Let the message signal $m(t) = A_m \cos(2\pi f_m t)$, where $A_m$is the peak amplitude of the modulating signal and $f_m$ is the modulating signal frequency. Substituting for $m(t)$ in equation (7.2.1), the equation for the FM signal is,

$$f_{FM} = f_c + K_f A_m \cos(2\pi f_m t)$$

$$f_{FM} = f_c + \Delta f \cos(2\pi f_m t) \tag{7.2.2}$$

where $\Delta f$ is the frequency deviation. It signifies the amount by which the carrier frequency gets deviated.

Multiplying by $2\pi$ on both sides of equation (7.2.2)

$$2\pi f_{FM} = 2\pi f_c + 2\pi\Delta f \cos(2\pi f_m t)$$

$$\Delta\omega_{FM} = \Delta\omega_c + \Delta\omega \cos(2\pi f_m t) \tag{7.2.3}$$

Since, $\omega_{FM} = \frac{d\theta(t)}{dt}$, integrating both sides of equation (7.2.3)

$$\omega_{FM}(t) = \omega_c(t) + \Delta\omega \int \cos(2\pi f_m t)\, dt \tag{7.2.4}$$

$$\theta(t) = \omega_c(t) + {}^{\Delta\omega}/_{\omega_m} \sin(2\pi f_m t)$$ (since $\omega_{FM}(t) = \theta(t)$ ) (7.2.5)

Therefore, the equation of FM Signal is given by,

$$V_{FM}(t) = A_c \cos[\theta(t)]$$

$$= A_c \cos[\omega_c t + {}^{\Delta\omega}/_{\omega_m} \sin(2\pi f_m t)]$$

$$= A_c \cos[\omega_c t + {}^{2\pi\Delta f}/_{2\pi f_m} \sin(2\pi f_m t)]$$

$$= A_c \cos[\omega_c t + {}^{\Delta f}/_{f_m} \sin(2\pi f_m t)]$$

$$= A_c \cos[\omega_c t + \beta \sin(2\pi f_m t)] \tag{7.2.6}$$

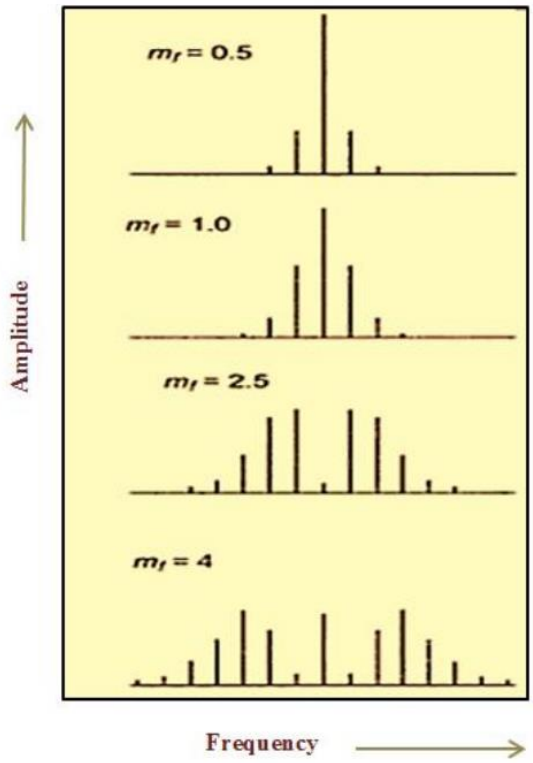
Where $m_f = \beta = \frac{\Delta f}{f_m}$ is defined as modulation index of FM. Unlike AM modulation index, $m_f$ is not restricted to one. It can be more than unity.

• **Spectrum of FM**

Figure 7.2.2 shows the spectrum of FM for different values of $m_f$. It is seen that as the modulation index, $m_f$ increases, more number of sidebands appear. Therefore any FM signal with large $m_f$ will have a large number of sidebands and hence larger bandwidth. Ideally, FM signal has infinite bandwidth. However, for practical purpose, Carson's rule is followed, which says that for good reception of FM, it is enough if those many side bands which constitutes 98% of power is taken. This acts as the basis for estimation of bandwidth for FM.

The bandwidth for FM as per Carson's rule is given by,y

$$Bandwidth \approx 2(\Delta f + f_m) \tag{7.2.7}$$




*Self -test:*

1 Define frequency modulation.
2 Write the time domain expression for frequency modulation and explain.
3 Define modulation index and write the expression for the same.
4 List the factors that affect the bandwidth of FM signal?

***Example Problems:***

1 Given a FM equation $V_{FM}(t) = 10 \cos[2\pi 10^8 t + 5\sin(2\pi\, 15000t)]$, calculate carrier frequency, modulating frequency, frequency deviation and bandwidth.

Solution:

Carrier frequency: $f_c = 10^8$Hz

Modulating frequency: $f_m = 15$ kHz

Frequency deviation: $\Delta f = \beta \times f_m = 5 * 15 = 75$ kHz

Bandwidth $= 2(\Delta f + f_m) = 2(75 + 15) = 180$ kHz

2 In an FM system when the audio frequency is 500Hz , modulating voltage is 2.5 , the deviation produced is 5kHz. If the modulating voltage is now increased to 7.5 calculate the new value of frequency deviation. If the AF voltage is raised to 10 while the modulating frequency is dropped to 250Hz, what is the frequency deviati produced. Also calculate modulation index in each case.

Solution:

Given: $f_m = 500$Hz, $A_m = 2.5$V, $\Delta f = 5 \times 10^3$Hz.

i) Modulation index: $\beta = \dfrac{\Delta f}{f_m} = \dfrac{5\times10^3}{500} = 50$

If $A_m = 7.5$V, $\Delta f =$ ?

$$K_f = \frac{\Delta f}{A_m} = \frac{5\times10^3}{2.5} = 2 \text{ kHz /V}$$

$$\Delta f = K_f \times A_m = 2 \times 7.5 = 15 \text{ kHz}$$

$$\text{Modulation index: } \beta = \frac{\Delta f}{f_m} = \frac{15\times10^3}{500} = 30$$

ii) $\Delta f = K_f \times A_m = 2 \times 10 = 20$ kHz

iii) Modulation index: $\beta = 80$

**Exercises**

1 A carrier of amplitude 5V and frequency 90MHz is frequency modulated by a sinusoidal voltage of amplitude 5V and frequency 15 KHz. The frequency sensitivity is 1Hz/V. Calculate the frequency deviation and modulation index. (Ans: $\Delta f = 5$Hz, $\beta=0.0003$)

2 The carrier frequency in an FM modulator is 1000 KHz. If the modulating frequency is 15 KHz, what are the first three upper sideband and lower sideband frequencies?

(Ans: 955kHz, 970kHz, 985kHz, 1015kHz,1030kHz,1045kHz)

.

### *7.2.3 Comparison of AM and FM*

| Sl. no | Parameter | AM | FM |
|---|---|---|---|
| 1 | Amplitude of the modulated wave | Varies instantaneously with the modulating signal amplitude | constant |
| 2 | Frequency of the modulated wave | Contains Carrier and sideband frequency components | Contains carrier and infinite sideband frequency components |
| 3 | Modulation Index | $0 < m_a \leq 1$ | $m_f > 0$ |
| 4 | Noise immunity | Less | More |
| 5 | Adjacent channel interference | More | Less due to guard bands |
| 6 | Bandwidth | Less | More |
| 7 | Circuit complexity | Less | More |
| 8 | Coverage area | More | Less |

*Self -test:*

1. Noise interference in AM is greater than in FM (True/False)
2. AM systems require more bandwidth than FM(True/False)
3. FM signals reach longer distances than AM (True/False)
4. AM systems are more complex to build than FM (True/False)

Exercises

1 Explain why FM waves cover shorter distances as compared to AM?

2 What is guard band? Explain.

Watch this Video for animation of amplitude modulation:

1. http://www.youtube.com/watch?v=_5JyiFWLn-w

Watch this Video for animation of frequency modulation:

1. http://www.youtube.com/watch?v=SmW4z76KgNQ

**Summary**

In this module we have learnt:

1. The definition of Frequency Modulation is a process of altering the frequency of the carrier signal with respect to the instantaneous amplitude of the modulating signal.
2. To draw the waveforms for frequency modulated signal with respect to the chosen modulating and carrier signals.
3. Modulation index gives the depth of modulation and is given by $m_f = \beta = \frac{\Delta f}{f_m}$
4. The bandwidth is estimated approximately by Carlson Rule given by Bandwidth = $2(\Delta f + f_m)$

# Chapter-8

# Introduction to digital communication

In analog communication, the message signals are continuous in nature and can take infinite amplitude levels. When these signals are transmitted over long distances, even a small disturbance (noise) can cause distortion in the signal. Once the signal is distorted, the noise cannot be removed and as a result the original signal cannot be recovered perfectly at the receiver. With digital techniques, these disturbances can be removed by detecting and correcting the errors. Hence digital communication has more benefits as compared to analog communication. In digital communication, the message signal is in discrete form with finite amplitude levels. If the message signal is analog, it must be converted to digital form by the process of Analog-to-Digital Conversion.

**Module-1: Digitization of Analog signals**

For any analog information to be transmitted using digital communication system, the signal must be converted to digital form. The Analog-to-Digital conversion process comprises of sampling, quantizing and encoding the analog signal. The digitized signal is then modulated using digital modulation techniques.

**Learning Outcomes:**

At the end of this module, students will be able to:

1. State Nyquist Sampling Theorem and explain.
2. Explain qualitatively pulse amplitude modulation technique with the help of suitable waveforms.
3. Draw and explain the general block diagram of Digital Communication system.
4. Explain qualitatively different types of digital modulation techniques with the help of suitable waveforms.

### 8.1.1 Basic principle of Sampling

The basic principles of Analog to Digital Conversion are as shown in Fig 8.1.1.Sampling of analog signals is the first step used to digitize analog information. Sampling can be observed in numerous real life applications. For example, music CDs (Compact Discs) are produced by sampling music signal at frequent intervals followed by quantizing and encoding each sample. Even in digital photography, periodic snapshots (samples) are taken to capture continuous phenomena. If the sampling rate is fast enough, the human sensory organs cannot discern the gaps between each snapshot when they are played back. This is the principle behind motion pictures. If the sampling rate is not fast enough, there will be distortion in the reconstructed picture obtained from the digitized samples. Therefore, while sampling an analog signal, there is a minimum sampling rate requirement, called the Nyquist Sampling rate that avoids distortion in the reconstructed signal. Harry Nyquist proved the sampling theorem which states that "It is possible to reconstruct a band-limited analog signal from periodic samples, as long as the sampling rate ( fs)is at least twice the highest frequency component (fm)of the signal". Mmathematicaly it can be expressed as

$$\mathbf{fs \geq 2fm} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (8.1)$$

Where **fs** is the sampling frequency and **fm** is highest frequency component in the signal. This theorem is also commonly called the Nyquist criteria for sampling or Sampling theorm.

This means that , for example if a voice signal has frequencies ranging from 0 to 4kHz (Low pass signal), then according to the Nyquist Sampling Theorem, in order to sample this signal without distortion, the minimum required sampling rate is equal to 8kHz. If an analog signal has frequency components ranging from 2 kHz to 5 kHz (Band pass signal), then according to the sampling theorem, the Nyquist sampling rate is equal to twice that of the bandwidth of the signal.ie. 2*(5-2) kHz = 6 kHz and not 10 kHz.

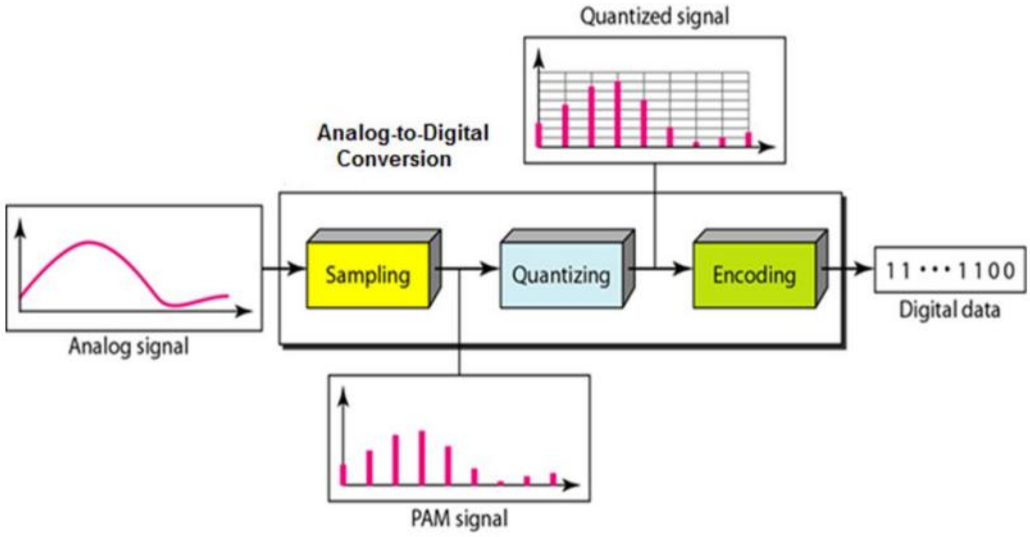



Fig 8.1.1 Steps involved in Analog-to Digital Conversion

***Self -test:***

1. *If a* signal is composed of frequency components up to 10 kHz., what is the minimum frequency at which the signal should be sampled as per Nyquiest criteria?

***Example Problem:***

1. Consider the analog signal x (t) =3cos100 $\pi$ t. Determine the minimum sampling rate required to avoid aliasing.
   Solution:
   The frequency of the analog signal can be calculated as $2\pi fc=100\pi$.
   Therefore fc=50Hz.According to the Nyquist sampling rate, the minimum sampling rate required to avoid aliasing is fs = 100 Hz.

***Exercise:***

1. **Consider the analog signal x (t) = 3cos50$\pi$t+10sin300$\pi$t-cos100$\pi$t. What is the Nyquist rate of sampling for this signal? (Ans: 300Hz)**

***Note:** The effect of incorrect sampling rate can be seen when the rotation of a helicopter blade is observed. As the speed of the blade increases, our eyes are under sampling the true speed of the blade with a rate which is limited by the human brain. Similarly, in movies, when the motion of car wheels with increasing speed is observed, the movie camera is under sampling the motion of car wheels by sampling at a rate equal to the fixed frames per second of the camera. In both the examples it is observed that as the speed increases, it creates an illusion of backward rotation. This is because in both cases the actual speed is under sampled.*

For Analogy of sampling to Wagon wheel effect, visit the following link: http://www.youtube.com/watch?v=6XwgbHjRo30

**Introduction to pulse modulation techniques**

Pulse modulation involves communication of information using a train of pulses as carrier. It may be used to transmit either analog information such as continuous speech or digital data. If the modulating signal is in analog form, it is called analog pulse modulation technique. Some of analog pulse modulation techniques are: Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM) and Pulse Position Modulation (PPM). If digital data is used to modulate a train of pulses, then it is called digital pulse modulation technique. Some of the digital pulse modulation techniques are: Pulse Code Modulation (PCM) and Delta Modulation (DM).

Analog Pulse modulation is a process in which continuous waveforms are sampled at regular intervals using a train of recurrent pulses. Information regarding the signal is transmitted only at the time of sampling along with any synchronizing pulses that may be required. At the receiving end, the original waveforms can be reconstructed from such samples, if the samples are taken as per the sampling theorem or Nyquist criteria . In analog pulse modulation, the sample amplitude may be infinitely variable while in digital pulse modulation such as PCM and DM, a code which indicates the sample amplitude that is assigned the nearest predetermined discrete amplitude level is sent.

A pulse train has three parameters, namely, Pulse Amplitude, Pulse Width and the instant of occurrence of the pulse – Pulse Position. The information to be transmitted can be used to vary any of these parameters according to the instantaneous amplitude of the modulating signal. This leads to three different types of pulse modulation and they are Pulse Amplitude Modulation, Pulse Width Modulation and Pulse Position Modulation as shown in Fig 8.1.2.

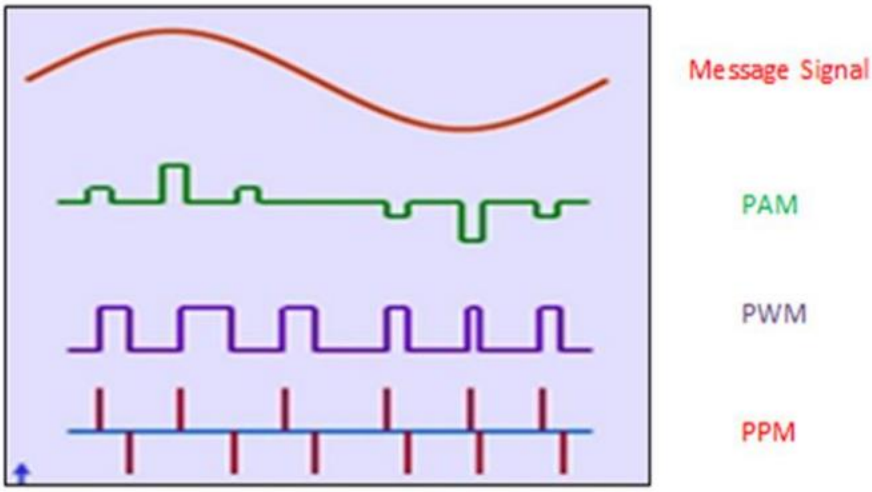



Fig 8.1.2: Different types of pulse modulation signals

### 8.1.2 Pulse Amplitude Modulation (PAM)

Pulse Amplitude Modulation (PAM) is the simplest form of pulse modulation as shown in Fig 8.1.3. Here the signal is sampled at regular intervals and each sample is made proportional to the amplitude of the message signal at the instant of sampling.

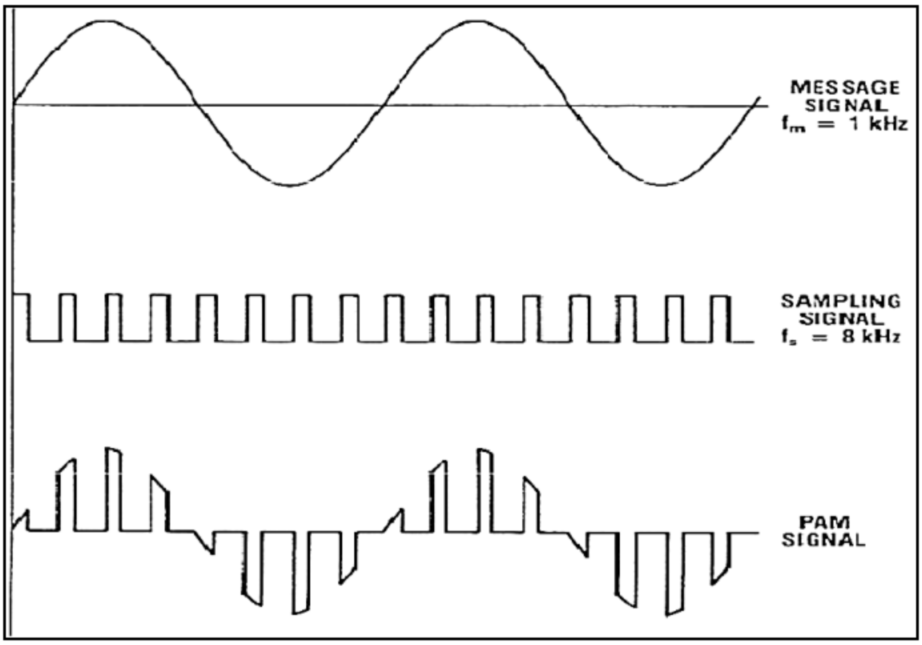



Fig 8.1.3: PAM signal

**8.1.3 Pulse Width Modulation**

The Pulse Width Modulation is also called Pulse Duration Modulation (PDM). Here the amplitude and starting time of each pulse is fixed but the width of each pulse is made proportional to the amplitude of the modulating analog signal at that instant (Refer Fig 8.1.2). PWM has lot of real world applications. For example, the transmission of voice or music can be performed by using PWM. Pulse Width Modulatio is also used in dimmer circuits to control the intensity of light. PWM is also used in many control applicationssuch as control of speed and torque of DC motors etc.,

**8.1.4 Pulse Position Modulation (PPM)**

In this method, both the amplitude and the duration are kept constant while the position of each pulse in relation to the position of a recurrent reference pulse is varied by each instantaneous sampled value of the modulating signal (Refer Fig 8.1.2). PPM is used in both analog and digital data transmission. It is commonly used in optical fibre communication, remote controls for TV, toys etc.

**8.1.5 Digital Communication System**

The pulse modulation techniques discussed in the previous section are used to transmit message signals over short distances. They are also called baseband modulation techniques. If message signals have to be transmitted over longer distances, then the pulse modulation technique is not suitable because the modulated signal is in digital form (information is contained in either the amplitude or width or position of the train of pulses). For this reason, digital modulation techniques is used, also called band pass modulation techniques. Here a continuous signal such as a high frequency sinusoidal signal acts as carrier. Digital modulation is achieved by varying either the amplitude or frequency or phase of the carrier in accordance with the digital data to be transmitted.

***ELEMENTS OF DIGITAL COMMUNICATION SYSTEMS:***

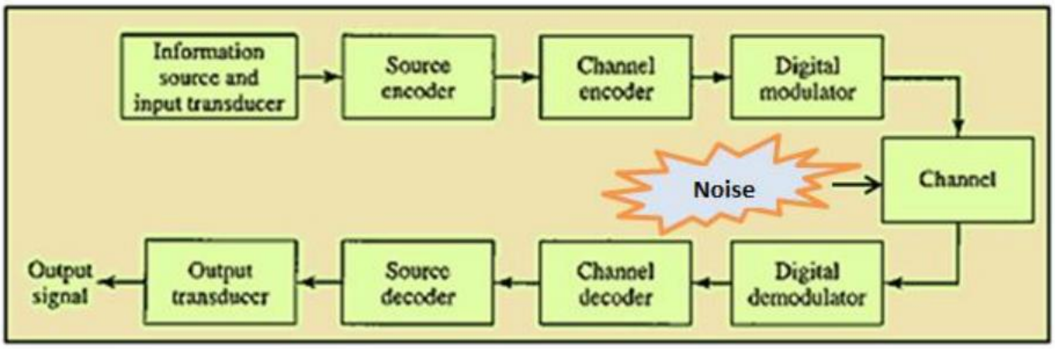



Fig 8.1.4 Block Diagram of a Digital Communication System

Fig 8.1.4 shows the functional elements of a digital communication system. The function of each block is explained as follows:

**a. Information Source and Input transducer.**

Many of the real world signals are physical in nature. The device used to convert these physical parameters to corresponding electrical signals is called input transducer. Examples of physical parameters are voice, speech, image etc. The input transducer used to convert voice, speech or music signal or image to an electrical signal. Examples of input transducers are microphone, camera etc. Usually, the output signal from the transducer will be analog in nature. This analog signal is converted into digital form by using analog –to-digital converter. The analog- to-digital conversion consists of sampling, quantizing and encoding. In the case of the output data of a computer, the signal is available in digital form directly.

**b. Source Encoder / Decoder**

The aim of the source coding is to represent the digital signal efficiently with as much less number of bits as possible. This will reduce the bandwidth required for transmission. Ex: Huffman coding. The source decoder performs the inverse operation of source encoder. ie. It is used to get back the data in its original representation.

**c. Channel Encoder / Decoder**

Channel coding consists of systematically adding extra bits in a known manner to the digital data to be transmitted. These extra bits do not convey any information but help the receiver to detect and / or correct some of the errors in the received data. Channel encoding is done by using either Block Coding or Convolution Coding methods. The channel decoder performs the inverse operation of channel encoder. ie. It is used to extract the digital data from its encoded form with minimum possible error. The decoder helps in detecting and/or correcting the errors in the received data that gets introduced during transmission.

**d. Modulator/ Demodulator**

The Modulator converts the input digital information into an electrical waveform suitable for transmission over the communication channel. Mainly there are three types of Digital modulation techniques viz., Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Phase Shift Keying (PSK).The extraction of the digital data from the received signal is accomplished by the demodulator.

**e. Channel**

The channels are either wired such as pair of wires, coaxial cable and optical fiber or wireless (free space) such as radio channel, satellite channel or combination of any of these. The communication channels have only finite bandwidth and the signal often suffers amplitude and phase distortion as it travels over the channel in addition to attenuation of signal. It may also get corrupted by unwanted, unpredictable electrical signals referred to as noise. The two important parameters used to measure the channel characteristics are Signal to Noise power Ratio (SNR) and usable bandwidth.

- **Advantages of Digital Communication**

1. The effect of distortion, noise and interference is less in a digital communication system. This is because the disturbance must be large enough to change the pulse from logic 0 to logic 1 or vice versa.

2. Regenerative repeaters can be used at fixed distance along the link, to identify and regenerate a pulse before it is degraded to an ambiguous state.

3. Digital circuits are more reliable and cheaper compared to analog circuits.4. The Hardware implementation is more flexible than analog hardware because of the use of microprocessors, VLSI chips etc.

5. Signal processing functions like encryption, compression can be employed to maintain the secrecy of the information.

6. Error detecting and Error correcting codes improve the system performance by reducing the probability of error

- **Disadvantages of Digital Communication**
    1. Large System Bandwidth: - Digital transmission requires a large system bandwidth to communicate the same information in a digital format as compared to analog format.

2. System Synchronization: - Digital detection requires system synchronization whereas the analog signals generally have no such requirement.

**8.1.6 Digital Modulation Techniques**

There are three basic types of digital modulation techniques. They are:

i. Amplitude Shift Keying (ASK)
ii. Frequency Shift Keying (FSK)
iii. Phase Shift Keying (PSK)

In all these techniques, amplitude, frequency or phase of a sinusoidal carrier is varied to represent the information which is to be sent. Here the digitized information is mapped into any one of the above three aspects of sine wave and then transmitted. The sine wave at the receiver is remapped back to the information. The digital modulation techniques are widely used in MODEMs ( MOdulator DEModulator) , mobile communication etc. Usually, FSK and PSK modulations are more frequently used than ASK.

**i. Amplitude Shift Keying (ASK)**

Here the amplitude of the carrier is changed in accordance with the binary information to be transmitted while the frequency and phase of the carrier are kept fixed. Bit 1 is transmitted by a carrier of one particular amplitude. Bit 0 is transmitted by changing the amplitude to 0 Volt or no signal in this case. The ASK also called On-Off keying (OOK) is as shown in Fig 8.1.5.

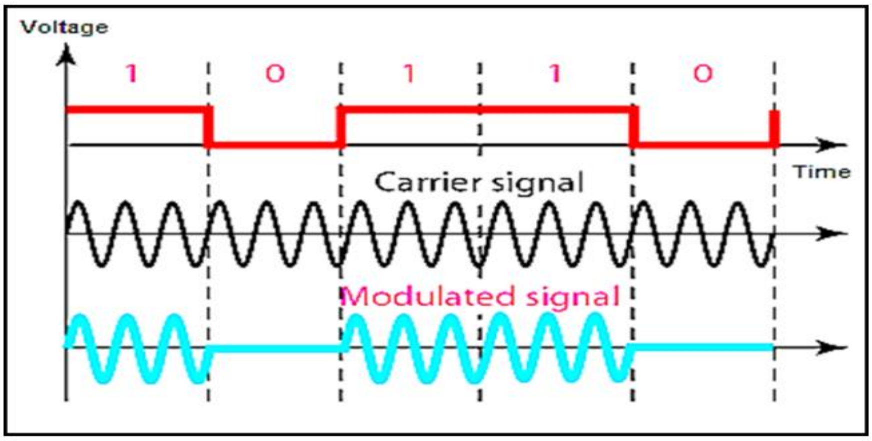



Fig 8.1.5 Binary ASK signal

**ii. Frequency Shift Keying (FSK)**

Here the frequency of the carrier is changed in accordance with the binary information. ie. one frequency for bit 1 and another frequency for bit 0. FSK signal is as shown in Fig8.1.6.

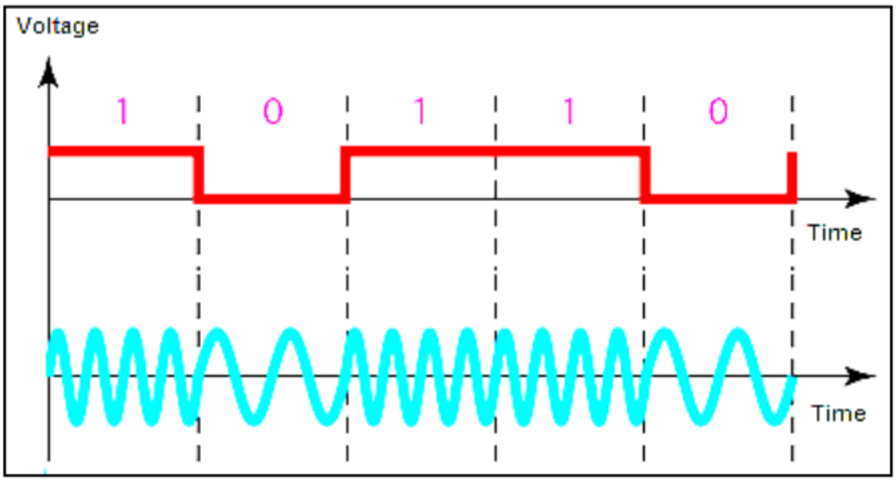



Fig 8.1.6 Binary FSK signal

**iii. Phase Shift Keying (PSK)**

Here the phase of the sinusoidal carrier is changed in accordance with the binary information. Phase in this context is the starting angle at which the sinusoid starts. To transmit bit 0, the phase of the sinusoid is shifted by 180o whereas to transmit bit 1, the phase of the sinusoidal carrier is shifted by another 180o . Thus the carrier phase shift represents the change in the state of the information. Fig 8.1.7 shows the binary PSK representation.

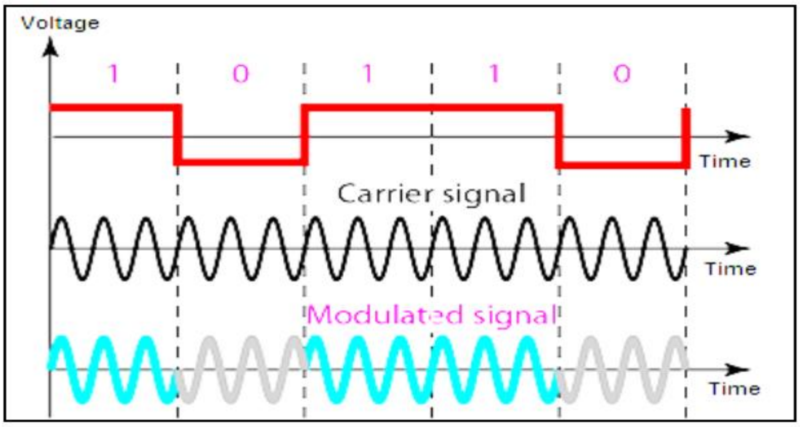



Fig 8.1.7 Binary PSK Signal

**Summary**

1. Sampling theorem states that "It is possible to reconstruct a band-limited analog signal from periodic samples, as long as the sampling rate is at least twice the highest frequency component of the signal".
2. Pulse modulation involves communication of information using a train of pulses as carrier.
3. Some of analog pulse modulation techniques are: Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM) and Pulse Position Modulation (PPM).
4. Digital modulation is achieved by varying either the amplitude or frequency or phase of the carrier in accordance with the digital data to be transmitted.
5. There are three basic types of digital modulation techniques. They are:
   Amplitude Shift Keying (ASK)
   Frequency Shift Keying (FSK)
   Phase Shift Keying (PSK)